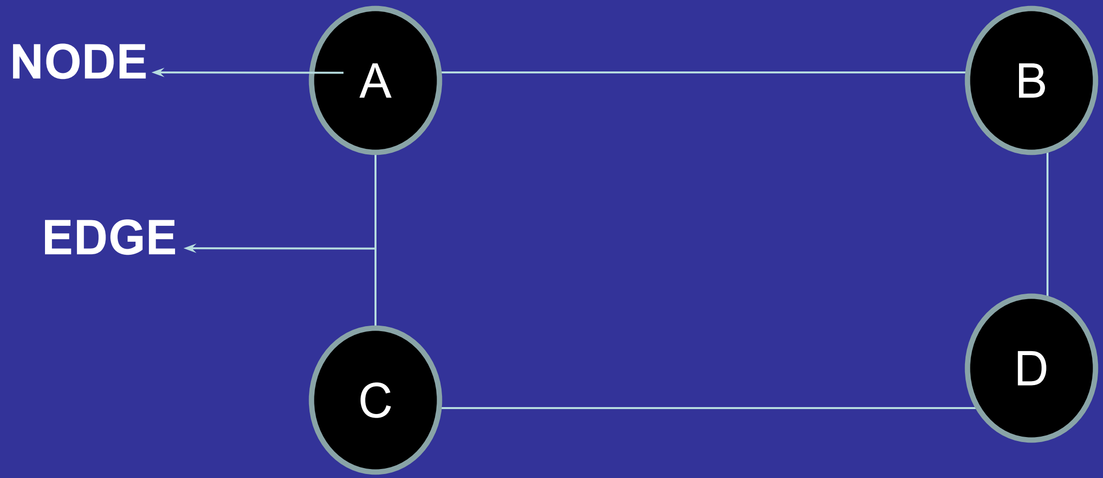
Graph

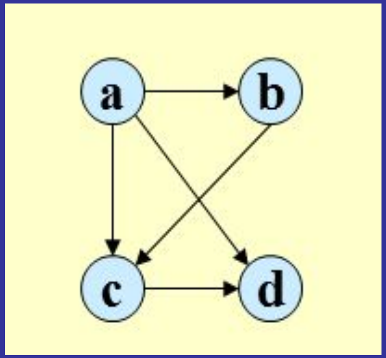
Data sometimes contain a relationship between pairs of elements which is not necessarily hierarchical in nature. The data structure which reflects this type of relationship is called a graph



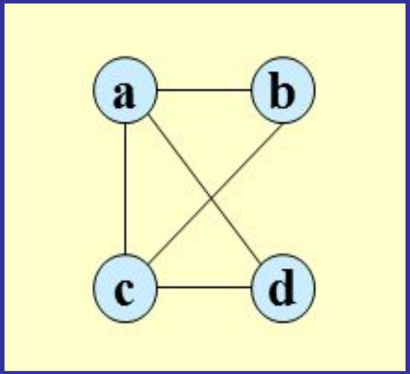
Classification of Graph

- Directed Graph
- Undirected Graph
- Weighted Graph
- Non-Weighted Graph

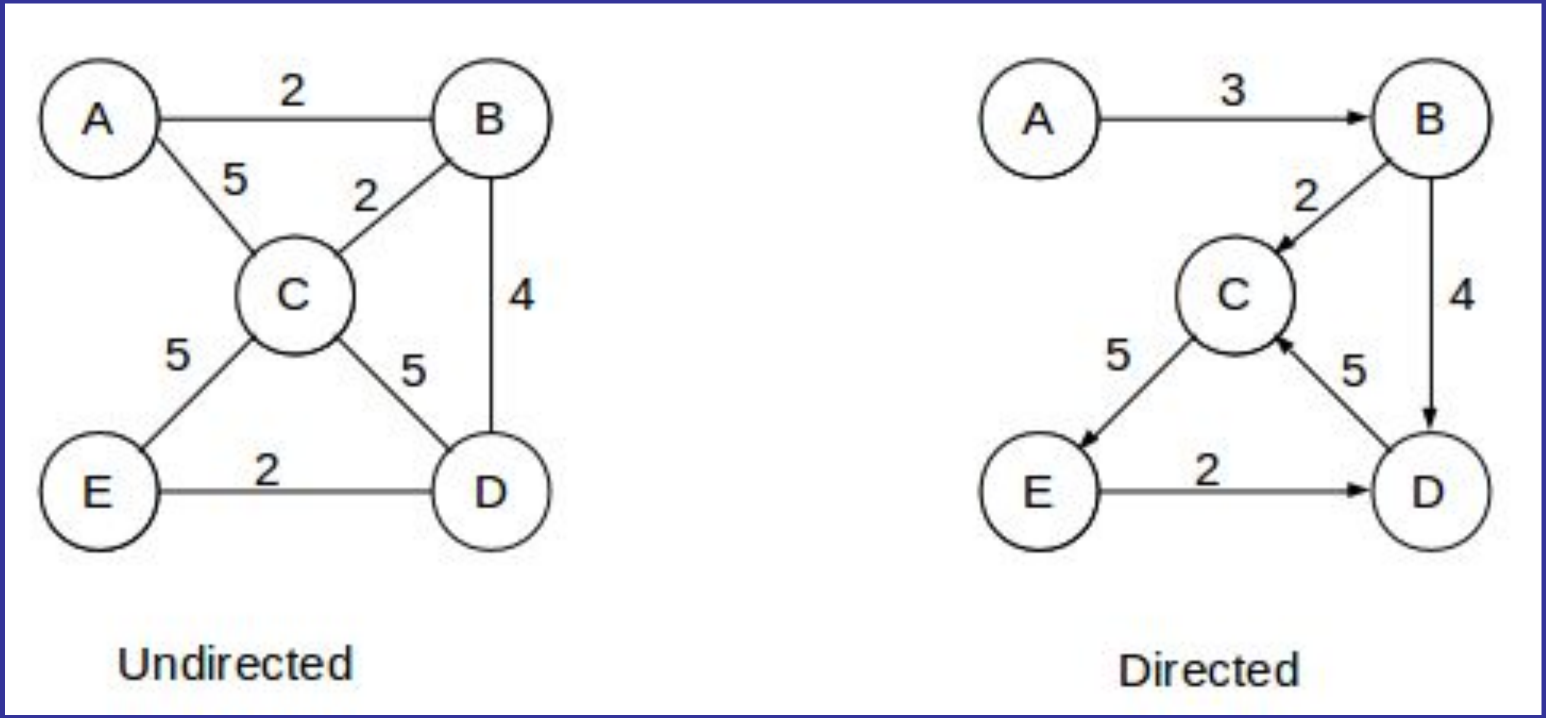
Directed graph contains edges with arrows.



Undirected graph contains edges without arrows.

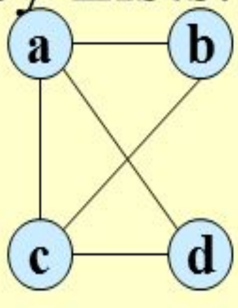


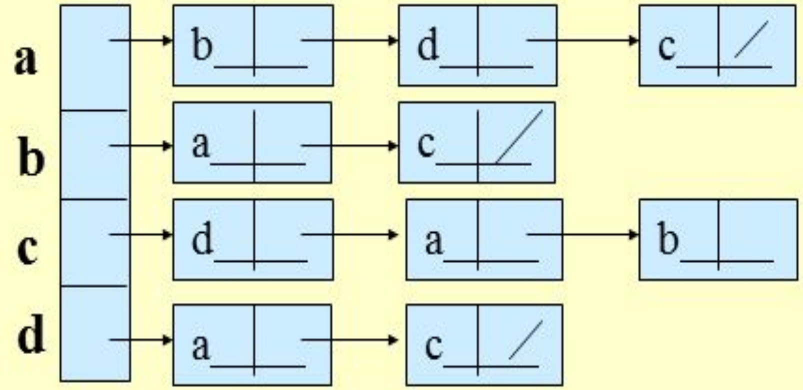
A weighted graph is a graph in which each branch is given a numerical weight.



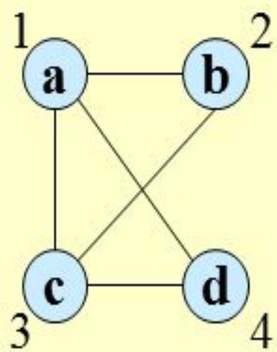
Representation of Graphs

- Two standard ways.
  - » Adjacency Lists.

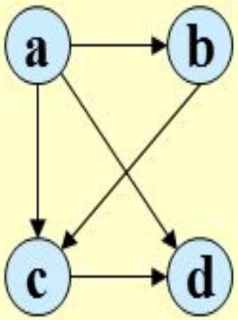


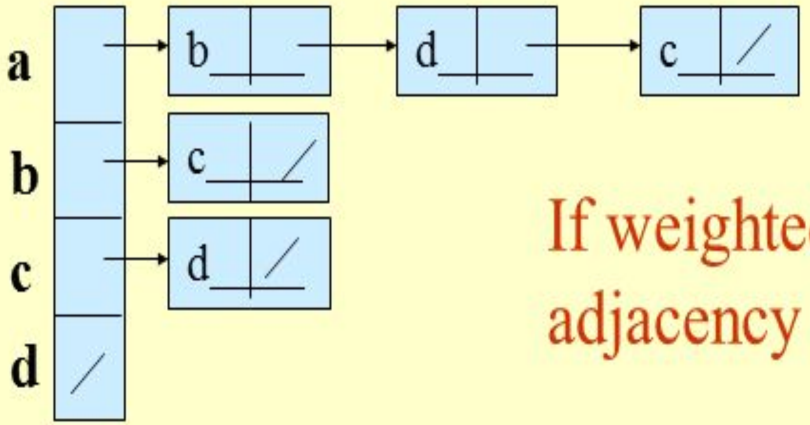


  - » Adjacency Matrix.

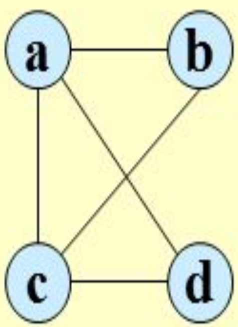


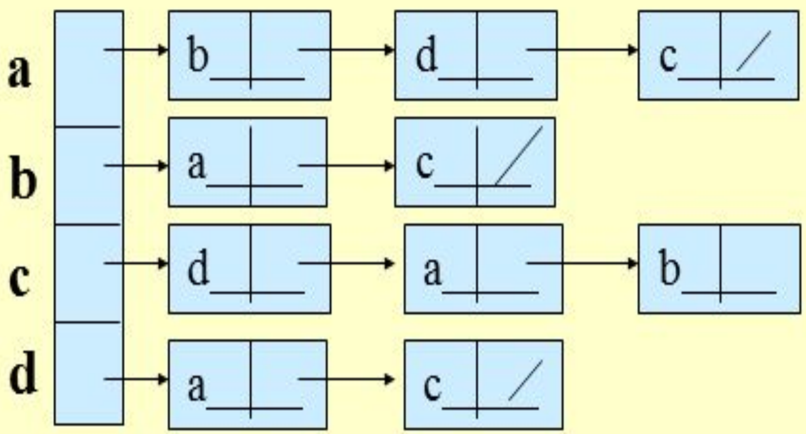
	1	2	3	4
1	0	1	1	1
2	1	0	1	0
3	1	1	0	1
4	1	0	1	0

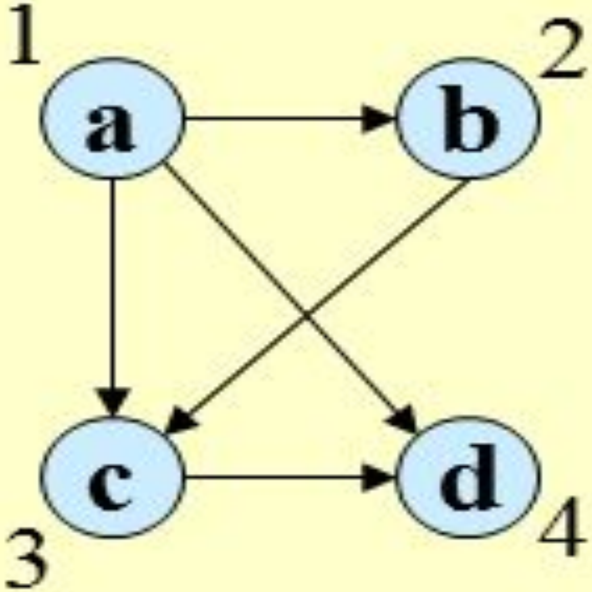




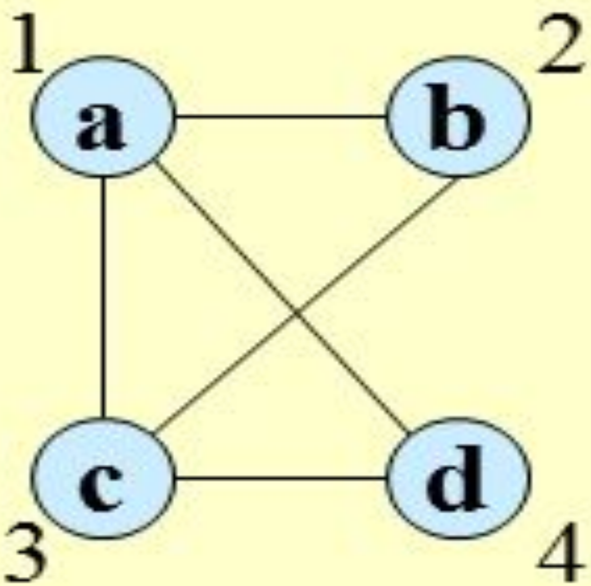
If weighted, store weights also in adjacency lists.







	1	2	3	4
1	0	1	1	1
2	0	0	1	0
3	0	0	0	1
4	0	0	0	0



	1	2	3	4
1	0	1	1	1
2	1	0	1	0
3	1	1	0	1
4	1	0	1	0